

Route-F P54: Fixed-VEV Tadpoles, the Full Complex Doublet Matrix, and the Pati–Salam Matching Interface

P54PQ-v2 four-dimensional mainline

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Abstract

We fix a symmetry-preserving finite tadpole prescription and compute every entry of the bosonic 4×4 complex doublet curvature, retaining the real and imaginary copy mixing. At the historical tree background the retuned matrix has one zero eigenvalue and three positive heavy eigenvalues. Its light vector rotates by 0.0329814 radians. The coarse heavy-gap norm test fails, but an exact relative-matrix certificate proves positivity at the stated truncation. Neither higher-loop control nor all-sector quantum stability follows.

The subsequent matching audit finds an independent error in the historical P2 interface: its PS beta table and threshold projectors do not describe the same active spectrum. The $SU(2)_L$ matching logarithm must have slope -71 , whereas the historical implementation has zero slope. We derive a same-field-content repair of all six logarithmic identities. Finite matching, the complete P54 Yukawa evolution and a physical constrained global fit remain uncomputed; no fit or model exclusion is fabricated.

Contents

1	Frozen action, background and scope	2
2	Step 1: an invariant fixed-VEV tadpole prescription	2
3	Step 2: the complete complex doublet matrix	3
4	Step 3: the corrected bosonic light state	4
5	Fermionic completion in the same tadpole prescription	5
6	Canonical Spin(10) normalization: a second independent repair	6
7	Type-II matching requires the full mixed triplet inverse	7
8	Step 4 audit: the Pati–Salam interface must be repaired	7
9	A constrained fit, not an unconstrained matrix optimization	9
10	Reproducibility and precise completion boundary	10

1 Frozen action, background and scope

The action remains P54PQ-v2, with fields $3\,16_F + 54_{H,\mathbb{R}} + 126_{H,\mathbb{C}} + 10_{H,\mathbb{C}} + 1_{H,\mathbb{C}}$. We denote them by $\Psi, \Phi, \Sigma, \phi, S$. Their PQ charges are $(-1, 0, +2, -2, -4)$, and the two symmetric family matrices enter as

$$-\mathcal{L}_Y = \frac{1}{2}(Y_{10})_{ij}(\Psi_i\Psi_j)_{10}\phi^* + \frac{1}{2}(Y_{126})_{ij}(\Psi_i\Psi_j)_{\overline{126}}\Sigma + \text{h.c.} \quad (1)$$

Scalar conjugation in these two terms differs. The scalar implementation uses 328 canonical real coordinates, with complex coefficients $(x_R + ix_I)/\sqrt{2}$. The chosen background is

$$\Phi_0 = \omega \text{diag}(-\frac{2}{5}I_6, \frac{3}{5}I_4), \quad (2)$$

$$\Sigma_0 = \sigma \Omega_{126}, \quad S_0 = v_s/\sqrt{2}, \quad \phi_0 = 0, \quad (3)$$

$$\sigma/\omega = 0.1265, \quad v_s/\omega = 0.25, \quad g_U = 0.562602889985. \quad (4)$$

We use $\omega = 1$ for numerical tensors and $\mu = g_U\omega$ for the *value* of the subtraction scale. All field derivatives hold μ and renormalized couplings fixed. The historical scale $M_U = 8.5347508424 \times 10^{14}$ GeV is not recertified here: Section 8 finds a defect in its matching calculation.

The bosonic hard functional is the previously declared background-field Landau- $\overline{\text{MS}}$ functional,

$$V_{1,B}^{\text{hard}} = \frac{1}{64\pi^2} \{ \text{Tr}_{290} F_{3/2}(H(x)) + 3 \text{Tr}_{33} F_{5/6}(M_V^2(x)) \}, \quad F_c(t) = t^2[\log(t/\mu^2) - c]. \quad (5)$$

Here $H = V_0''$ is the scalar operator and $(M_V^2)_{AB} = g_U^2(T_A x)^T(T_B x)$. The 38 soft scalar and 12 soft vector modes are excluded by a gapped spectral branch, not by differentiating a fixed diagonal list. The vector coefficient includes the gauge/ghost prescription of this hard potential. This is a zero-momentum matching curvature, not a pole mass and not a gauge-independent observable.

2 Step 1: an invariant fixed-VEV tadpole prescription

2.1 Why exactly three radial conditions suffice at this vacuum

Let $R = (\partial_\omega x_0, \partial_\sigma x_0, \partial_{v_s} x_0)$. The canonical metric gives

$$R^T R = \text{diag}(12/5, 2, 1). \quad (6)$$

The SM-singlet tangent has five real dimensions: one in 54, a complex singlet in 126, and the complex S . Its two phase directions are the broken abelian gauge orbit and the PQ orbit. They span the two phase tangents because the gauge orbit acts on Σ_0 and PQ also acts nontrivially on S_0 . There is no additional physical singlet phase at this background. An SM-invariant gradient has no component in nontrivial SM representations; invariance under the two phase orbits removes the other two singlet components. Thus only the three radial tadpoles remain. This argument uses a symmetry-preserving regulator and the perturbative potential; it does not erase the nonperturbative PQ anomaly.

Write $t_\alpha = \partial_\alpha V_1^{\text{hard}}(x_0)$. Choose the finite counterterm within the already present invariant quadratic operators:

$$\delta V = -\frac{\delta\mu^2}{2}\Phi_{ij}\Phi_{ij} - \frac{\delta\nu^2}{2 \cdot 5!}\Sigma_{ijklm}\Sigma_{ijklm}^* - \delta\mu_s^2 S^* S. \quad (7)$$

Restricted to the vacuum,

$$\delta V(x_0) = -\frac{6}{5}\delta\mu^2\omega^2 - \frac{1}{2}\delta\nu^2\sigma^2 - \frac{1}{2}\delta\mu_s^2 v_s^2. \quad (8)$$

Solving $\partial_\alpha(V_1 + \delta V) = 0$ gives

$$\boxed{\delta\mu^2 = \frac{5t_\omega}{12\omega}, \quad \delta\nu^2 = \frac{t_\sigma}{\sigma}, \quad \delta\mu_s^2 = \frac{t_s}{v_s}.} \quad (9)$$

This defines the finite mass-renormalization convention at the chosen scale. It must be used also in subsequent thresholds; it is not an additional change in field content.

The canonical real Hessian is

$$H_{\text{ct}} = -\delta\mu^2 P_{54} - \frac{1}{2}\delta\nu^2 P_{126} - \delta\mu_s^2 P_S. \quad (10)$$

The factor $1/2$ for 126 follows from the declared kinetic/tensor normalization. Replacing it by one fails the phase Ward identity. For an antisymmetric generator T , invariance implies $\nabla V(x)^T T x = 0$. Differentiation gives

$$H(x)Tx = T\nabla V(x). \quad (11)$$

Consequently the loop curvature along a broken orbit need not vanish before tadpole subtraction. At the fixed stationary point, $(\Pi + H_{\text{ct}})Tx_0 = 0$. The two independent normalized phase directions provide a sensitive test of the counterterm normalization.

2.2 Numerical tadpoles and Ward checks

The bosonic calculation gives

$$(t_\omega, t_\sigma, t_s)/\omega^3 = (-0.135193703018, -0.691922627373, -0.008559428597), \quad (12)$$

$$(\delta\mu^2, \delta\nu^2, \delta\mu_s^2)/\omega^2 = (-0.056330709591, -5.469744089909, -0.034237714387). \quad (13)$$

For the normalized Σ phase, the loop/CT curvatures are -2.734872044954 and $+2.734872044955$; for the S phase they are -0.034237714387 and $+0.034237714387$. Their residuals are below $1.3 \times 10^{-13}\omega^2$. Phase and doublet tadpoles vanish to the checked precision. The full SM-singlet projector equals the projector onto the three radial and two phase directions.

The raw subtraction $|\delta\nu^2|/|\nu^2| \simeq 14.32$ is large. Its cancellation against a common multiplet curvature is required by the Ward identity; the ratio is scheme dependent, not by itself a proof of breakdown. Nevertheless it demands a genuine higher-loop/naturalness assessment. We do not infer perturbative convergence from a successful one-loop stationarity test.

3 Step 2: the complete complex doublet matrix

3.1 Canonical complex copies and exact spectral derivatives

Let B be a 328×4 complex orthonormal basis in the complexification of the real scalar representation with

$$C_3 = 0, \quad C_2 = 3/4, \quad iR_Y = 1/2, \quad iR_{T_3} = -1/2. \quad (14)$$

Each of the geometric sectors $(\phi_{\text{hol}}, \phi_{\text{anti}}, \Sigma_{\text{hol}}, \Sigma_{\text{anti}})$ has rank one here. Define $Q_R = \sqrt{2}\Re B$, $Q_I = -\sqrt{2}\Im B$. Then (Q_R, Q_I) is an orthonormal real eight-plane and $e^{-\pi R_Y} Q_R = Q_I$. A Hermitian complex mass matrix D has the real form

$$D_{\mathbb{R}} = \begin{pmatrix} \Re D & -\Im D \\ \Im D & \Re D \end{pmatrix}. \quad (15)$$

We compute all four diagonal, six real mixed and six imaginary mixed entries. Imaginary jets are also tested independently. The phase of each column is fixed by its largest canonical coordinate being real positive; this is a scalar coordinate convention, not an inferred fermionic Clebsch phase convention.

For an operator A with eigenvalues λ_i , write $A_a = \partial_a A$, $A_{ab} = \partial_a \partial_b A$, in its eigenbasis. For a separated hard branch put $f'_i = F'(\lambda_i)$ on the hard modes and zero on soft modes. The exact second variation is

$$\partial_a \partial_b \text{Tr}_{\text{hard}} F(A) = \sum_i f'_i(A_{ab})_{ii} + \sum_{ij} f'^{[1]}_{ij}(A_a)_{ij}(A_b)_{ji}, \quad (16)$$

where $f'_{ij}^{[1]} = (f'_i - f'_j)/(\lambda_i - \lambda_j)$, with the continuous diagonal/degenerate limit. Hard-soft divided differences are retained. Dropping them would discard field-dependent eigenvector mixing. Since V_0 is quartic, $H(x)$ is quadratic, so centered first and second jets are exact polynomials up to floating-point roundoff. Steps 0.02ω and 0.01ω agree at better than $10^{-12}\omega^2$ for the independent curvature check.

3.2 Computed matrix, before and after the finite subtraction

In units of ω^2 , the tree matrix is

$$D_0 = \begin{pmatrix} .016799176 & .015202796 & 0 & .000326645 \\ .015202796 & .013758748 & 0 & 0 \\ 0 & 0 & .285626100 & -.010606602 \\ .000326645 & 0 & -.010606602 & .138405400 \end{pmatrix}. \quad (17)$$

The complete hard bosonic correction is

$$\Pi_B = \begin{pmatrix} -.0194382305 & -.0002813614 & -.0000003330 & .0002014259 \\ -.0002813614 & -.0213868984 & .0000302589 & -.0000020155 \\ -.0000003330 & .0000302589 & -2.5883375274 & .0000609813 \\ .0002014259 & -.0000020155 & .0000609813 & -2.6730027447 \end{pmatrix}. \quad (18)$$

The imaginary entries have magnitude below 2.8×10^{-13} for this real-coupling benchmark; they were computed, not set to zero. Separate scalar/vector matrices and the full canonical basis are exported in the JSON ledger. The old tree-light projection $c_0^\dagger \Pi_B c_0 = -0.0202366510950\omega^2$ is reproduced.

On this four-copy space,

$$D_{\text{ct}} = -\frac{\delta\nu^2}{2}\text{diag}(0, 0, 1, 1), \quad P_{10} = \text{diag}(1, 1, 0, 0). \quad (19)$$

Thus the large common 126 curvature cannot be omitted while keeping the fixed VEVs. The retuned bosonic matrix is

$$D_B = D_0 + \Pi_B + D_{\text{ct}} - \delta\xi_{02}P_{10}. \quad (20)$$

The 10 has zero VEV, so this last finite mass tuning does not alter the three tadpole equations.

4 Step 3: the corrected bosonic light state

4.1 Schur equation and numerical eigenpair

In the basis (c_0, Q) , with c_0 the tree zero mode and Q its orthonormal complement, decompose

$$D_B = \begin{pmatrix} A & b^\dagger \\ b & C \end{pmatrix}. \quad (21)$$

For $C > 0$, solving the lower block gives the exact algebraic criterion

$$\boxed{A = b^\dagger C^{-1} b, \quad c_B = \frac{c_0 - Q C^{-1} b}{\sqrt{1 + b^\dagger C^{-2} b}}}. \quad (22)$$

Here “exact” refers to the assembled one-loop-truncated matrix, not to an all-orders field theory. Diagonalizing it resums some higher-order algebraic terms but supplies none of the missing two-loop diagrams.

The solution has $\delta\xi_{02}/\omega^2 = -0.0202627041913$ and

$$D_B/\omega^2 \simeq \begin{pmatrix} .0176236497 & .0149214344 & -.0000003330 & .0005280705 \\ .0149214344 & .0126345543 & .0000302589 & -.0000020155 \\ -.0000003330 & .0000302589 & .4321606176 & -.0105456204 \\ .0005280705 & -.0000020155 & -.0105456204 & .2002747003 \end{pmatrix}. \quad (23)$$

Its eigenvalues and phase-fixed light vector are

$$\text{spec}(D_B)/\omega^2 = (-1.1 \times 10^{-19}, 0.03025666325, 0.19979763559, 0.43263922302), \quad (24)$$

$$c_B \simeq (0.6462011668, -0.7631652015, 0.00001218358, -0.0017108974)^T. \quad (25)$$

The rotation from c_0 is 0.0329814289 radians, $\|b\|_2 = 0.000998486637\omega^2$, and both sides of the Schur condition are $0.00003290941346\omega^2$ with residual $6.2 \times 10^{-18}\omega^2$. Projecting the fixed-VEV correction alone would instead give $\delta\xi_{02}/\omega^2 = -0.0202297946947$.

4.2 A less wasteful, exact heavy-positivity certificate

The old sufficient norm test gives

$$\left\| Q^\dagger (D_B - D_0) Q \right\|_2 / \omega^2 = 0.1465345611 > \Delta_{\text{tree}}/\omega^2 = 0.0305570303. \quad (26)$$

Thus that test fails by a factor 4.79545; it may not be reported as passed. Its failure does not prove a negative eigenvalue, because it compares corrections in all directions with only the smallest mass. Let $C_0 = Q^\dagger D_0 Q > 0$ and

$$E = C_0^{-1/2} (C - C_0) C_0^{-1/2}. \quad (27)$$

By congruence,

$$C = C_0^{1/2} (I + E) C_0^{1/2} > 0 \iff \lambda_{\min}(E) > -1. \quad (28)$$

Numerically $\text{spec}(E) \simeq (-0.0109064, 0.4393632, 0.5234555)$, so $C \succeq 0.9890936C_0 > 0$. This proves positivity of this heavy block without treating the 0.1465 correction as a correction to the lightest heavy state. The largest relative correction is still about 52%; neither this certificate nor the positive spectrum establishes perturbative convergence or positivity of the other scalar sectors.

The corrected magnitudes give $|r| = 1.18100$, $|s| = 0.00602976$ under the previous scalar-only convention map, while $w_{126} = |c_3|^2 + |c_4|^2 = 2.92732 \times 10^{-6}$. Most of the rotation is within the 10 pair. A sizable total rotation is therefore not itself a flavor repair or a growth of the needed 126 component. Absolute spinor normalization and PS transport must precede a physical flavor inference.

5 Fermionic completion in the same tadpole prescription

For a fixed invertible symmetric $M_R = \sigma f_M$, define $X = M_R^\dagger M_R$. Here f_M denotes the Majorana-normalized matrix; it is not silently identified with the unit-bidoublet coupling defined below. The hard three-Majorana functional at vanishing electroweak background is

$$V_{1,F}^{\text{hard}} = -\frac{1}{32\pi^2} \text{Tr} \left[X^2 \left(\log \frac{X}{\mu^2} - \frac{3}{2} \right) \right]. \quad (29)$$

Since $\partial_\sigma X = 2X/\sigma$, spectral functional calculus gives

$$t_{\sigma,F} = -\frac{1}{8\pi^2\sigma} \text{Tr} \left[X^2 \left(\log \frac{X}{\mu^2} - 1 \right) \right], \quad \delta\nu_F^2 = \frac{t_{\sigma,F}}{\sigma}. \quad (30)$$

In this minimal unbroken-electroweak sector $t_{\omega,F} = t_{s,F} = 0$. This is a partial field derivative; μ must not vary along with ω even though its chosen numerical value is $g_U\omega$.

For doublet-dependent $D(z) = \sum_a z_a Y_a$ and the mass matrix

$$\mathcal{M}(z) = \begin{pmatrix} 0 & D(z) \\ D(z)^T & M_R \end{pmatrix}, \quad (31)$$

the previously derived full complex Hessian is

$$(\Pi_F)_{ab} = -\frac{1}{8\pi^2} \text{Tr}(Y_a^\dagger Y_b W), \quad W = X \left(\log \frac{X}{\mu^2} - 1 \right). \quad (32)$$

Its use in the fixed-VEV prescription requires

$$\Delta D_F = \Pi_F - \frac{\delta \nu_F^2}{2} P_{126}. \quad (33)$$

One must combine this matrix with (20) and solve (22) again. Adding only a fitted scalar number η_Y or only Π_F is insufficient. The Y_a and f_M must be derived from the same two underlying matrices used in the PS fit, with their canonical Clebsch projections. Synthetic tests verify this functional identity but do not supply fitted matrices. Three complex-family tests, including a degenerate Takagi pair and all eight real doublet directions, pass 11/11: the radial and Hessian relative errors are below 2.6×10^{-10} and 8.8×10^{-8} , respectively; family/copy unitary covariance also holds, with P_{126} transformed along with the copy basis.

6 Canonical Spin(10) normalization: a second independent repair

This calculation can be done before a fit. Let X, Y, Z be the Pauli matrices and set

$$\Gamma_{2k-1} = Z^{\otimes(k-1)} \otimes X \otimes I^{\otimes(5-k)}, \quad \Gamma_{2k} = Z^{\otimes(k-1)} \otimes Y \otimes I^{\otimes(5-k)}. \quad (34)$$

Then $\{\Gamma_i, \Gamma_j\} = 2\delta_{ij}$, $\Gamma_* = (-i)^5 \Gamma_1 \cdots \Gamma_{10}$ and $C = \Gamma_2 \Gamma_4 \Gamma_6 \Gamma_8 \Gamma_{10}$. The tensors $C\Gamma_i$ and $C\Gamma_{i_1 \dots i_5}$ are symmetric, while the three-form tensors are antisymmetric. Restrict to the chiral 16-dimensional subspace selected by the P1 self-duality. If U_{Ia} is the actual orthonormal five-form basis, define

$$(B_{126,a})_{AB} = \sum_{i_1 < \dots < i_5} U_{i_1 \dots i_5, a} (C\Gamma_{i_1} \cdots \Gamma_{i_5})_{AB}. \quad (35)$$

The ordered-index sum equals the action-card $1/5!$ contraction. For each of the 45 generators R , the verifier checks

$$R_{16}^T B_a + B_a R_{16} + \sum_b B_b (R_{126})_{ba} = 0, \quad (36)$$

and the analogous vector identity. Thus the normalization is linked to the actual scalar tensor, not guessed from another author's notation.

On the unit P1 singlet Ω_{126} the raw five-form contraction has rank one with sole singular value $4\sqrt{2}$. On a unit canonical doublet, the down-type raw coefficients are $\sqrt{2}$ for 10 and $2/\sqrt{3}$ for 126; the signed relative lepton/down Clebsch is -3 . Consequently, for the raw action-card matrices Y_{10}, Y_{126} ,

$$h_D = \sqrt{2}Y_{10}, \quad f_D = \frac{2}{\sqrt{3}}Y_{126}, \quad f_M = 4\sqrt{2}Y_{126} = 2\sqrt{6}f_D. \quad (37)$$

These positive CG normalization constants fix absolute magnitudes; the complex vertices must also carry a common scalar/spinor phase convention. With unit canonical scalar doublets and the P1 radial σ ,

$$M_R = 2\sqrt{6}\sigma f_D = \sigma f_M. \quad (38)$$

Alternatively, if one calls f_M simply f , the Dirac 126 vertex is $f/(2\sqrt{6})$. One cannot use both unit Dirac f and $M_R = \sigma f$ with the same canonical σ . The action card is corrected accordingly; the field content and number of independent family matrices do not change.

The selected Clifford chirality has conjugate-matter hypercharges in the P1 generator orientation. A global conjugation is allowed, but the scalar holomorphic/antiholomorphic map and fermion states must be conjugated together. The absolute factors above and the signed relative -3 are checked; they do not yet constitute the phase-resolved projection of all four scalar copies. Previous external-matrix CW numbers are historical conditional tests, not physical values after this correction. Since the heavy spectrum also enters a logarithm, a silent constant rescaling of the final η_Y number is not a valid repair.

The independent LL triplet contraction has raw magnitude $4\sqrt{2}$, so the Majorana-normalized convention has $|M_L| = |f_M \Delta_{\text{can}}|$. The actual exported $Y = +1$ triplet is a Σ -antiholomorphic direction in the selected conjugate-matter orientation; its conjugate has unit overlap with the required holomorphic LL tensor. This verifies the absolute triplet factor as well. The common complex phase convention with Y_ν and the light doublet still must be exported before combining type I and type II. All 20/20 Clifford, covariance, normalization and triplet-alignment checks pass.

7 Type-II matching requires the full mixed triplet inverse

The actual tree spectrum has two complex $(1, 3, |Y| = 1)$ copies from 54–126 mixing, with

$$m_{T,1}^2/\omega^2 = 0.2688312001, \quad m_{T,2}^2/\omega^2 = 0.5416068249.$$

The 54 also contains a real $Y = 0$ triplet, which is not the direct LL -coupled channel. Let e_A span all neutral real directions of the $|Y| = 1$ triplets, and let q be the unit real neutral light direction with $x = x_0 + vq$ corresponding to $H^0 = v/\sqrt{2}$. Set

$$K_{AB} = V_0^{(2)}[e_A, e_B], \quad J_A = V_0^{(3)}[e_A, q, q]. \quad (39)$$

Then

$$V = V_0 + \frac{1}{2}z^T K z + \frac{v^2}{2}J^T z + \dots, \quad z_{\text{ind}} = -\frac{v^2}{2}K^{-1}J + O(v^4/M_T^3). \quad (40)$$

Define the canonical fermionic contraction $\mathcal{Y}_A^{LL} = \partial M_{LL}/\partial z_A$. In the convention $m_\nu = v^2 C_5/2$,

$$\boxed{C_5^{II} = -\sum_{AB} \mathcal{Y}_A^{LL} (K^{-1})_{AB} J_B, \quad C_5^I = -Y_\nu M_R^{-1} Y_\nu^T.} \quad (41)$$

The fit uses $C_5 = C_5^I + C_5^{II}$, unless a controlled bound shows that one term is negligible. Even $J_{126} = 0$ would not suffice to set type II to zero: $(K^{-1})_{126,54} J_{54}$ can be nonzero. Representation/PQ allowance of a source is not a numerical computation of its Clebsch. The finite one-loop seesaw matching must be adapted only after the additional PS fields have been matched, rather than importing a pure SMEFT formula at the wrong stage [3, 4].

The new scalar-source calculation uses the exported complex B, c_B , with $q = \sqrt{2}\Re(Bc_B)$, and finds in the phase-fixed complex (54, 126) triplet basis

$$K_{\mathbb{C}}/\omega^2 = \begin{pmatrix} .541600225 & -.001341735117i \\ .001341735117i & .268837800 \end{pmatrix}. \quad (42)$$

In real order $(54_R, 126_R, 54_I, 126_I)$,

$$J/\omega \simeq (0, .002684241669, -.063294492019, 0)^T, \quad (43)$$

$$\omega z_{\text{ind}}/v^2 \simeq (0, -.004700734233, .058421207024, 0)^T. \quad (44)$$

The magnitude of the 126 response from its direct source is $0.004992368741/\omega$, while the 54 source contributes $0.000291634508/\omega$ with the opposite phase. Suppressing mass mixing would give $0.004992307015/\omega$: mixing reduces this by about 5.84%. The ten checks include closure of the complete triplet subspace, canonical real/complex equivalence, two derivative steps, positivity and the projected field equation. This uses tree K, V_0''' with a bosonic-improved q ; it is a mixed-order scalar diagnostic, not a loop-accurate type-II Wilson coefficient or a fitted neutrino matrix.

8 Step 4 audit: the Pati–Salam interface must be repaired

8.1 The active census and an exact beta-table discrepancy

For group order $(4, L, R)$, the declared four active complex scalar parents have the following indices and quadratic Casimirs:

Parent	d_C	(T_4, T_L, T_R)	(C_4, C_L, C_R)
$\phi : (1, 2, 2)$	4	(0, 1, 1)	(0, 3/4, 3/4)
$\Sigma : (15, 2, 2)$	60	(16, 15, 15)	(4, 3/4, 3/4)
$\Delta_L : (10, 3, 1)$	30	(9, 20, 0)	(9/2, 2, 0)
$\Delta_R : (\overline{10}, 1, 3)$	30	(9, 0, 20)	(9/2, 0, 2)

Using Weyl fermions and complex scalars, the gauge-and-matter terms are

$$a_i = -\frac{11}{3}C_i(G) + \frac{2}{3}\sum_F T_i(F) + \frac{1}{3}\sum_S T_i(S), \quad (45)$$

$$b_{ij} = -\frac{34}{3}C_i(G)^2\delta_{ij} + \sum_F [2C_j(F) + \frac{10}{3}C_i(G)\delta_{ij}]T_i(F) \\ + \sum_S [4C_j(S) + \frac{2}{3}C_i(G)\delta_{ij}]T_i(S). \quad (46)$$

For three 16_F families, $\sum_F T = (6, 6, 6)$. Exact rational evaluation then gives

$$a_{\text{PS}} = (2/3, 26/3, 26/3), \quad b_{\text{PS}} = \begin{pmatrix} 3551/6 & 249/2 & 249/2 \\ 1245/2 & 779/3 & 48 \\ 1245/2 & 48 & 779/3 \end{pmatrix}. \quad (47)$$

Yukawa-dependent two-loop gauge terms are additional and cannot be set to zero in a claimed full fit. The historical table has $a_4 = 1$ and $b_{44} = 1209/2$. Its difference is exactly one extra complex $(6, 1, 1)$, which contributes $\Delta a_4 = 1/3$, $\Delta b_{44} = 38/3$. The historical coefficients agree with Eq. (7) of [1], while the active census follows the two sixes assigned to M_U in its Table 1. Our independent index calculation exposes this discrepancy; agreement with a printed table is not independent validation.

8.2 A matching-scale failure independent of the extra six

The old two-site code assigns only Δ_R to the lower scalar threshold. Both that parent and the lower massive vectors are $SU(2)_L$ singlets. It therefore computes $\lambda_{I,L} \equiv 0$. But the same code uses the convention

$$\alpha_{2,\text{SM}}^{-1} = \alpha_{L,\text{PS}}^{-1} - \frac{\lambda_{I,L}}{12\pi}, \quad \frac{d\alpha_i^{-1}}{d\log\mu} = -\frac{a_i}{2\pi} + O(\alpha). \quad (48)$$

Differentiating demands

$$\boxed{\lambda'_{I,L} = -6(a_L^{\text{PS}} - a_2^{\text{SM}}) = -6(26/3 + 19/6) = -71,} \quad (49)$$

not zero. The residual matching equation changes by $71/(12\pi) = 1.88333349325$ per unit $\log\mu$ at one loop. This exact mismatch invalidates the physical matching-scale interpretation of the old root and covariance table. It is not an issue of a few decimal places and not a proof that the action has no viable point.

8.3 Same-content logarithmic repair in all six directions

Retain the four full parents above between the thresholds. In real-scalar index convention, $I_A = (68, 72, 72)$ and $I_{\text{full}} = (84, 84, 84)$. The PS-to-SM map, with SM order $(3, 2, 1)$, is

$$\mathcal{R}(v_4, v_L, v_R) = (v_4, v_L, \frac{2}{5}v_4 + \frac{3}{5}v_R), \quad I_h = (0, 1, 3/5). \quad (50)$$

Massive-vector indices follow from adjoint differences: $I_{V,U} = (4, 6, 6)$ and $I_{V,I} = (1, 0, 14/5)$. Subtract each vector site's Goldstone orbit once from the scalar census. The physical scalar indices are consequently

$$I_{S,U} = I_{\text{full}} - I_A - I_{V,U} = (12, 6, 6), \quad (51)$$

$$I_{S,I} = \mathcal{R}I_A - I_h - I_{V,I} = (67, 71, 67). \quad (52)$$

Physical real scalars contribute $+I_S \log(M/\mu)$ to λ and the vector/gauge/ghost sector contributes $-21I_V \log(M/\mu)$. Hence

$$\lambda'_U = 21I_{V,U} - I_{S,U} = (72, 120, 120), \quad (53)$$

$$\lambda'_I = 21I_{V,I} - I_{S,I} = (-46, -71, -41/5). \quad (54)$$

With $a_{10} = -34/3$, both vector identities hold exactly:

$$\lambda'_U = -6[a_{10}(1, 1, 1) - a_{\text{PS}}], \quad \lambda'_I = -6[\mathcal{R}a_{\text{PS}} - a_{\text{SM}}]. \quad (55)$$

This closes the six logarithmic counting checks. It does not compute the finite matching: at nonzero ω, σ , mass, parent and Goldstone projectors need not commute. Substituting the new projector into the old weighted mass logarithm is not a proof of a consistent two-stage finite EFT. One must match at a PS-preserving upper background and then in the lower EFT, or derive an equivalent covariant mixed-operator matching with the same gauge/tadpole prescription. No replacement M_I, M_U is claimed.

9 A constrained fit, not an unconstrained matrix optimization

9.1 Canonical Yukawa matching and scale covariance

At either threshold let $Y_a^{(0)} = \mathcal{P}_a(c, \mu)Y^+$ be the tree intertwiner/eigenvector projection. Hard full-minus-EFT vertices and kinetic matrices give, by expanding the inverse square roots of the three kinetic terms,

$$Y_a^- = Y_a^{(0)} + \Delta\Gamma_a^{\text{hard}} - \frac{1}{2} \left[(\Delta Z_L)^T Y_a^{(0)} + Y_a^{(0)} \Delta Z_R + \sum_b (\Delta Z_\phi)_{ba} Y_b^{(0)} \right] + O(\hbar^2). \quad (56)$$

The corresponding logarithmic check is

$$\frac{d\Delta Y}{d \log \mu} = \beta_Y^- - \mathcal{P}\beta_Y^+ - \dot{\mathcal{P}}Y^+ + O(\hbar^2). \quad (57)$$

The derivative of the doublet projection matters. The scalar number η_Y is not a replacement for vertex or fermion wave-function matching. Standard PS equations [2] are useful regression targets, but their minimal example retains Δ_R without Δ_L and has a different lower Higgs EFT. The full P54 left/right matrix beta functions and absolute canonical intertwiners must be supplied explicitly.

At nondegenerate type-I thresholds write $M_R = U^* \text{diag}(M_i) U^\dagger$ and $y_i = (Y_\nu U)_{.i}$. In the convention $m_\nu^I = -v^2 \kappa/2$, sequential removal gives $\kappa^- = \kappa^+ + y_i y_i^T / M_i$ at $\mu_i \simeq M_i$. Their order must follow the fitted Takagi masses. A common threshold M_I for all three is an extra approximation, not a convention.

9.2 Eliminate nuisance eigenvectors rather than fitting them freely

Let θ contain invariant scalar couplings, VEVs, gauge parameters and the two family matrices modulo common unitary family rotations. In a nonsingular Takagi patch one may take h diagonal nonnegative and retain complex symmetric f ; degenerate patches require their residual quotient. Define

$$\mathcal{D}(\theta, \delta\xi) = D_0 + \Pi_B + H_{\text{ct},B} + \Pi_F[h, f] + H_{\text{ct},F}[f] - \delta\xi P_{10}. \quad (58)$$

At each candidate, solve the tadpole equations and the unique light-state constraint, rather than allowing c and η_Y to move independently of the action. For a simple normalized eigenpair,

$$d\lambda = c^\dagger (d\mathcal{D})c, \quad Q_c d\mathcal{D} = -[Q_c(\mathcal{D} - \lambda)Q_c]^{-1} Q_c (d\mathcal{D})c, \quad c^\dagger dc = 0. \quad (59)$$

These are obtained by differentiating $\mathcal{D}c = \lambda c$ and projecting onto c and its complement. They supply implicit derivatives for a constrained optimizer; eigenvectors need not be refitted as free

spurions. In particular, $\partial\lambda/\partial\delta\xi = -c^\dagger P_{10}c < 0$ on this branch, so the light tuning is locally well posed as long as the gap remains open.

The physical likelihood is then

$$\chi_{\text{prof}}^2(\vartheta) = \inf_{\theta \in \mathcal{C}(\vartheta)} [\mathcal{O}_{\text{EFT}}(\theta) - d]^T \mathbf{C}^{-1} [\mathcal{O}_{\text{EFT}}(\theta) - d] + \chi_{\text{nuisance}}^2, \quad (60)$$

where $\mathcal{C}$ imposes same-action stationarity, exactly one light doublet, heavy positivity, a predeclared perturbativity/control criterion, consistent finite gauge/Yukawa matching, and both terms of (41). The quantity $\eta_Y = c^\dagger \Pi_{Fc}/\omega^2$ is a derived correlated nuisance once actual matrices are fitted, not an arbitrary independent knob. Experimental covariance and any threshold priors must be stated. An undefined EFT map is not an empty feasible domain; the optimizer must report “not ready”, not a spurious best fit or a proof of exclusion.

10 Reproducibility and precise completion boundary

The full-doublet implementation is `route_f/code/verify_p54_full_doublet_cw.py`, with JSON/Markdown outputs of the same stem. It exports the full canonical basis, separate scalar/vector matrices, radial counterterms, phase Ward residuals, retuned eigenpair and relative positivity certificate. Its 18 checks pass. The exact PS census is implemented in `route_f/code/verify_p54_ps_active_census.py`; its 14 audit tests pass while the two historical physical consistency gates fail. Historical P2 outputs are retained for reproducibility and explicitly superseded in the roadmap, not overwritten to conceal the discrepancy.

Further independent verifiers are `route_f/code/verify_p54_spinor_intertwiners.py` (20/20), `route_f/code/verify_p54_fermion_tadpole.py` (11/11), and `route_f/code/verify_p54_typeii_triplet_source.py` (10/10). The numerical implementation uses Python 3.9, NumPy 2.0.2, SciPy 1.13.1 and JAX/JAXLIB 0.4.30 in double precision. Expensive Hessians are cached under `tmp/p54_full_doublet_cw`, keyed by the P1 action source, parameters and exact field coordinates; an unchanged replay uses the same tensors instead of repeating a scan. Source hashes are exported.

Completed here: the invariant tadpole rule; numerical bosonic radial/phase tests; the complete bosonic complex doublet matrix; its corrected one-loop-truncated eigenpair; the exact six-direction PS logarithmic repair. The fermionic rule is a functional identity awaiting the same fitted matrices and canonical contractions. Still open: finite staged matching, complete P54 matrix RGEs, total quantum scalar stability/loop control, the self-consistent flavor/seesaw fit and physical pole observables. The independent soliton/GTA line does not block this work.

References

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- [2] D. Meloni, T. Ohlsson and S. Riad, *Renormalization Group Running of Fermion Observables in an Extended Non-Supersymmetric $SO(10)$ Model*, [arXiv:1612.07973](#).
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- [4] X. Li, D. Zhang and S. Zhou, *One-loop Matching of the Type-II Seesaw Model onto the Standard Model Effective Field Theory*, [arXiv:2201.05082](#).